

IMAT MOCK EXAM 8 — "Final Rehearsal"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 44/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Logical Reasoning & Reading (Q1-9)

1. "The human gut hosts trillions of microbes that assist digestion, synthesise vitamin K and several B vitamins, and help train the developing immune system. A course of antibiotics can transiently reduce the diversity of this community. In mice, transplanting gut microbes from obese donors transfers some of the donors' metabolic traits to lean recipients, and diet can reshape the community's composition within days."

Which of the following is not supported by the passage?

- A) Gut microbes influence their host's metabolism.
- B) Diet can change the composition of the gut community within days.
- C) Gut microbes synthesise some vitamins.
- D) Taking probiotic supplements cures inflammatory bowel disease.
- E) Antibiotics can temporarily reduce gut microbial diversity.

Answer: D

The passage explicitly states that gut microbes synthesise vitamin K and several B vitamins, that antibiotics transiently reduce diversity, that transplants transfer metabolic traits between mice, and that diet reshapes the community within days. It makes no claim anywhere about treating or curing inflammatory bowel disease with probiotic supplements, so that statement goes beyond the text. The probiotic claim is plausible-sounding health advice, but plausibility is not support: nothing in the passage links probiotics to any disease outcome.

2. A city finds that neighbourhoods with more street trees have lower childhood asthma rates. The council concludes that planting trees will reduce asthma.

Which of the following, if true, most weakens the council's conclusion?

- A) Tree-lined streets are concentrated in wealthier neighbourhoods, which also have better housing and healthcare.
- B) Asthma has many triggers.
- C) Trees produce pollen.
- D) Trees are planted on wide streets.
- E) Some neighbourhoods have parks rather than street trees.

Answer: A

The council infers causation from a correlation. If tree-lined streets are concentrated in wealthier neighbourhoods, then wealth plausibly drives both the tree planting and the lower asthma (better housing, healthcare, lower pollution exposure), so planting more trees need not change asthma rates at all: this confounder attacks the inference directly. By contrast, the fact that trees produce pollen, or that asthma has many triggers, does not explain why tree-rich areas show less asthma, so neither undermines the conclusion as strongly.

3. (General knowledge) The Milano Cortina 2026 Winter Olympic Games are the __ time that Italy has hosted the Winter Olympic Games.

- A) fourth B) first C) second D) fifth E) third

Answer: E

Italy has previously hosted the Winter Games at Cortina d'Ampezzo in 1956 and in Turin in 2006. Milano Cortina 2026 is therefore the third Italian Winter Olympics. Calling it the first or second would ignore one of those earlier editions.

4. A sealed box contains a number of levers. Exactly one of the following two statements is true:

- (1) All of the levers in the box are made of steel.
(2) None of the levers in the box is made of steel.

Which conclusion follows necessarily?

- A) No lever is made of steel.
B) The box contains at least one lever.
C) All the levers are made of steel.
D) Some, but not all, of the levers are made of steel.
E) The box is empty.

Answer: B

Step 1: test whether the box could be empty. If it were empty, both statements would be vacuously true (all zero levers are steel, and no zero levers are steel), contradicting the condition that exactly one is true. Step 2: so the box contains at least one lever. The condition does not tell us which statement is the true one: a non-empty box is consistent with either all steel or none steel. What follows necessarily is only the existence of at least one lever, and the claim that the box is empty is precisely the case ruled out.

5. Three tasks (triage, laboratory and records) must be assigned, one each, to Anil, Bea and Cara. Anil refuses to do triage. Cara refuses to do records.

How many different valid assignments are possible?

- A) 2 B) 3 C) 4 D) 5 E) 6

Answer: B

Count by Cara's allowed tasks (she refuses records, so she takes triage or laboratory). Case 1: Cara takes triage, then laboratory and records go freely to Anil and Bea, 2 ways. Case 2: Cara takes laboratory, then triage and records remain; Anil refuses triage, so Anil takes records and Bea takes triage, 1 way. Total $2 + 1 = 3$ valid assignments, all distinct people-to-task pairings, so the count is exact.

6. A clock reads 7:35. What is the angle between the hour hand and the minute hand?

- A) 12.5° B) 17.5° C) 20° D) 22.5° E) 35°

Answer: B

The minute hand moves 6 degrees per minute: at 35 minutes it sits at $35 \times 6 = 210$ degrees from 12. The hour hand moves 30 degrees per hour plus 0.5 degrees per minute: at 7:35 it sits at $7 \times 30 + 35 \times 0.5 = 210 + 17.5 = 227.5$ degrees.

The angle between them is $227.5 - 210 = 17.5$ degrees

The tempting 35 degree answer comes from comparing 210 degrees with $7 \times 30 = 210$ and forgetting that the hour hand has drifted 17.5 degrees past the 7.

7. In a dose-finding trial, patients rated their average pain on a 0 to 10 scale. At doses of 0, 2.5, 5 and 10 mg the mean pain scores were 8, 5, 3 and 3 respectively.

Which conclusion is best supported?

- A) Higher doses are always better.
- B) The drug is unsafe at 10 mg.
- C) Increasing the dose beyond 5 mg produces no further average reduction in pain.
- D) The drug has no effect on pain.
- E) 10 mg is the optimal dose.

Answer: C

The mean pain score falls from 8 to 5 to 3 as the dose rises from 0 to 5 mg, then stays at 3 at 10 mg. The data therefore support a plateau: increasing the dose beyond 5 mg produces no further average reduction. The trial measured only pain, so it says nothing about safety at 10 mg and cannot identify an optimal dose; the drug clearly does affect pain (8 down to 3), and higher doses are not always better because the score stops falling.

8. A share price rises by 10% one day and falls by 10% the next day. The net change over the two days is:

- A) a 1% decrease B) no change C) a 1% increase D) a 10% decrease E) a 19% decrease

Answer: A

Take a starting price of 100.

After the rise it is 110; after the fall it is $110 \times 0.90 = 99$

The net change is 1 percent down, because successive percentage changes multiply: $1.10 \times 0.90 = 0.99$. The answers of no change, or a 20 percent style swing, come from adding the percentages instead of applying them to the new base.

9. In a class of 30 students, 18 study biology, 15 study chemistry, and 8 study both. How many study neither subject?

- A) 2 B) 3 C) 4 D) 5 E) 6

Answer: D

By inclusion-exclusion, the number studying at least one subject is $18 + 15 - 8 = 25$ (the 8 who take both would otherwise be counted twice).

The number studying neither is $30 - 25 = 5$

Section B - Biology (Q10-32)

10. A father is blood group AB; a mother is blood group O (genotype OO). Which statement about their children is correct?

- A) All children are AB.
- B) The four ABO phenotypes are all equally possible.
- C) All children are O.
- D) The children's groups cannot be predicted.
- E) Each child must be A or B (genotypes AO or BO); AB and O are impossible.

Answer: E

The AB father passes either the A or the B allele to each child; the OO mother can pass only O. Every child is therefore AO (group A) or BO (group B), each with probability one half. No child can be AB (that needs A and B together) or O (that needs O from both parents), so exactly these two outcomes are impossible.

11. Genes A and B lie on the same chromosome and crossing over between them never occurs. An individual with chromosomes AB/ab (coupling arrangement) is crossed with ab/ab. The offspring will be:

- A)** only the two recombinant types
- B)** offspring in a 9:3:3:1 ratio
- C)** only the two parental types, AB/ab and ab/ab, in equal proportions
- D)** all offspring AB/ab
- E)** all four phenotypic classes in a 1:1:1:1 ratio

Answer: C

With no crossing over, the AB/ab parent produces only two gamete types, AB and ab, in equal numbers; the ab/ab tester produces only ab. The offspring are AB/ab and ab/ab in a 1:1 ratio, both combinations identical to the parents' arrangements. The classic 1:1:1:1 testcross ratio would require the recombinant gametes Ab and aB, which cannot arise without crossing over.

12. A woman is a carrier of haemophilia ($X^H X^h$). Her husband has haemophilia ($X^h Y$). What fraction of their daughters is expected to have haemophilia?

- A)** 0 **B)** 1/4 **C)** all of them **D)** 1/2 **E)** 3/4

Answer: D

Every daughter receives the father's X carrying the haemophilia allele. She is affected only if the mother also passes her marked X, an event with probability one half per daughter; the other half of daughters are carriers. So the expected fraction of affected daughters is 1/2. A quarter would be the fraction among all children, not among daughters specifically.

13. In double-stranded DNA, guanine pairs with cytosine through:

- A)** three hydrogen bonds
- B)** one hydrogen bond
- C)** four hydrogen bonds
- D)** a covalent bond
- E)** two hydrogen bonds

Answer: A

The decisive fact: a guanine-cytosine pair is joined by three hydrogen bonds, while an adenine-thymine pair has only two. This is why GC-rich DNA needs more heat to separate. The bonds are hydrogen bonds, not covalent links between the strands.

14. Retroviruses such as HIV copy their RNA genome into DNA inside the host cell using:

- A)** DNA polymerase of the host
- B)** restriction enzymes
- C)** ribozymes
- D)** RNA polymerase
- E)** reverse transcriptase

Answer: E

Retroviruses carry reverse transcriptase, an RNA-dependent DNA polymerase that copies the viral RNA genome into DNA, which then integrates into the host chromosome. The host's own DNA polymerase copies DNA templates, not RNA, and RNA polymerase makes RNA from DNA, the wrong direction for this step.

15. During translation, the mRNA codon 5'-AUG-3' is read by the tRNA anticodon. Written in the 3' to 5' direction, the anticodon is:

- A)** 3'-UAC-5' **B)** 3'-AUG-5' **C)** 3'-TAC-5' **D)** 5'-UAC-3' **E)** 5'-ATG-3'

Answer: A

Codon and anticodon pair antiparallel. Reading the codon 5'-AUG-3', the pairing runs A with U, U with A, G with C, so the anticodon written in the 3' to 5' direction is 3'-UAC-5'. Written in the 5' to 3' direction the same anticodon would read CAU, and any sequence containing T cannot be right because tRNA is RNA.

16. Which protein exhibits quaternary structure?

- A)** ribonuclease, a single 124-residue chain
B) pepsin, a single-chain enzyme
C) haemoglobin, with two α and two β chains
D) myoglobin, a single 153-residue chain
E) an α -helix

Answer: C

Quaternary structure means an assembly of more than one polypeptide chain. Haemoglobin, with its four subunits (two alpha and two beta chains), is the textbook example. Myoglobin, pepsin and ribonuclease are each a single chain and so show at most tertiary structure, while an alpha-helix is a motif of secondary structure, not a whole protein organisation.

17. When two amino acids join to form a dipeptide, the reaction is:

- A)** hydrolysis, consuming water
B) a condensation reaction releasing one molecule of water and forming a peptide bond
C) a substitution of the R groups
D) a neutralisation releasing salt
E) an oxidation releasing hydrogen

Answer: B

The amino group of one amino acid attacks the carboxyl group of the other; one molecule of water is released and a peptide (amide) bond forms. This is a condensation reaction. Hydrolysis is the reverse process, breaking the bond by adding water, so the option describing hydrolysis with water consumed gets the direction of the reaction backwards.

18. Fermentation yields only about 2 ATP per glucose, versus roughly 30 by full aerobic oxidation, because:

- A)** NAD^+ is never regenerated in fermentation
B) the electron transport chain runs backwards during fermentation
C) glycolysis stops when fermentation begins
D) fermentation produces no ATP at all
E) the glucose is only partially oxidised: most of its energy remains trapped in lactate or ethanol

Answer: E

Fermentation is glycolysis plus the regeneration of NAD^+ , and it stops at reduced end-products such as lactate or ethanol. The glucose carbon is never fully oxidised to CO_2 and water, and the Krebs cycle and electron transport chain never run, so almost all of the glucose's energy remains locked in the end-products. The 2 ATP that are made all come from glycolysis, so fermentation certainly does produce ATP.

19. Substrate-level phosphorylation (direct transfer of a phosphate from a substrate to ADP) occurs in:
- A) pyruvate oxidation only
 - B) the electron transport chain only
 - C) both glycolysis and the Krebs cycle
 - D) glycolysis only
 - E) the Krebs cycle only

Answer: C

Substrate-level phosphorylation is direct transfer of a phosphate group from a substrate to ADP. It occurs at two steps of glycolysis (the phosphoglycerate kinase and pyruvate kinase steps) and at one step of the Krebs cycle (succinyl-CoA synthetase). The electron transport chain instead uses oxidative phosphorylation through a proton gradient, which is why the option naming it is wrong.

20. Leaves look green because chlorophyll:
- A) is a green pigment that absorbs all wavelengths equally
 - B) destroys green light
 - C) absorbs only green light
 - D) absorbs red and blue light strongly and reflects/transmits green light
 - E) emits green fluorescence under sunlight

Answer: D

Chlorophyll's absorption peaks strongly in the blue and red regions of the spectrum; green light is largely reflected and transmitted, so that is the light reaching our eyes and leaves look green. The claim that it absorbs only green light inverts the truth (leaves would then appear the complementary colour), and absorbing all wavelengths equally would make leaves look black.

21. The sodium-potassium pump of the cell membrane moves:
- A) 3 Na⁺ out for every 2 K⁺ in, at the cost of one ATP
 - B) 2 Na⁺ out for every 3 K⁺ in
 - C) Na⁺ and K⁺ both down their gradients
 - D) 3 K⁺ out for every 2 Na⁺ in
 - E) equal numbers of Na⁺ and K⁺ in opposite directions

Answer: A

The sodium-potassium pump (Na⁺/K⁺-ATPase) hydrolyses one ATP to export 3 Na⁺ and import 2 K⁺, each ion moved against its gradient. This electrogenic imbalance (net loss of one positive charge per cycle) contributes to the resting potential and keeps the ionic gradients that nerve and muscle firing consume. The ratio 3 out : 2 in at a cost of one ATP is the only version matching the textbook pump.

22. Which checkpoint ensures that DNA replication is complete and undamaged before mitosis begins?
- A) the M checkpoint (spindle)
 - B) the G2 checkpoint
 - C) the S checkpoint
 - D) cytokinesis
 - E) the G1 checkpoint

Answer: B

The G2 checkpoint sits after S phase and before mitosis, and it verifies that DNA replication is complete and that any damage is repaired before the cell commits to division. The G1 checkpoint decides whether to begin a new cycle (cell size, nutrients, DNA integrity at the start), and the spindle checkpoint acts inside mitosis at metaphase, too late to check replication.

23. Meiosis II resembles mitosis because in both processes:

- A)** the cell becomes haploid for the first time
- B)** bivalents align on the equator
- C)** sister chromatids separate and move to opposite poles
- D)** homologous chromosomes pair and cross over
- E)** the chromosome number is halved

Answer: C

In both meiosis II and mitosis, sister chromatids separate and move to opposite poles, so the mechanics of segregation are the same. Pairing of homologues, crossing over, halving of the chromosome number and alignment of bivalents on the equator are all features of meiosis I, and the cell first becomes haploid at the end of meiosis I, already before meiosis II begins.

24. The inside of a resting neurone is negative relative to the outside mainly because:

- A)** the membrane is impermeable to all ions at rest
- B)** chloride ions enter the cell continuously
- C)** large anions are actively transported out
- D)** potassium ions diffuse out through leak channels down their gradient, leaving unbalanced negative charge inside
- E)** sodium ions are pumped into the cell

Answer: D

At rest the membrane is most permeable to potassium through open K⁺ leak channels. Potassium diffuses out down its concentration gradient, leaving behind unbalanced intracellular negative charge (large fixed anions that cannot follow), and this builds the resting potential of about minus 70 mV. The pump maintains the gradients over time, but its 3-out : 2-in imbalance is only a minor direct contributor to the voltage.

25. Which sequence correctly traces a simple reflex arc?

- A)** receptor → CNS → effector → motor neurone → sensory neurone
- B)** effector → motor neurone → CNS → sensory neurone → receptor
- C)** receptor → motor neurone → CNS → sensory neurone → effector
- D)** CNS → receptor → sensory neurone → motor neurone → effector
- E)** receptor → sensory neurone → CNS → motor neurone → effector

Answer: E

A reflex arc runs: receptor detects the stimulus, sensory (afferent) neurone carries the signal to the CNS, integration occurs in the centre, motor (efferent) neurone carries the command outward, and the effector (muscle or gland) responds. Every sequence that puts the motor neurone before the CNS, or the CNS before the receptor, reverses the direction of information flow.

26. Which hormone raises blood glucose as part of the acute stress response?

- A)** ADH (vasopressin)
- B)** insulin
- C)** oxytocin
- D)** aldosterone
- E)** adrenaline (epinephrine)

Answer: E

The decisive fact: adrenaline, released from the adrenal medulla as part of the acute fight-or-flight response, stimulates glycogenolysis in the liver and raises blood glucose to fuel rapid action. Insulin lowers blood glucose (the opposite direction), ADH and aldosterone regulate water and sodium balance rather than glucose, and oxytocin concerns labour and milk ejection.

27. Vaccination with a live attenuated (weakened) pathogen produces which type of immunity?
- A) active artificial immunity, with the person's own plasma cells and memory cells
 - B) natural passive immunity via the placenta
 - C) natural active immunity after catching the disease
 - D) passive artificial immunity from the injected antigens themselves
 - E) no immunity, since the pathogen is weakened

Answer: A

A live attenuated vaccine delivers an artificially introduced but weakened microbe that multiplies slightly and provokes a genuine primary immune response, generating the person's own plasma cells and memory cells: active, artificially acquired immunity. Passive immunity would mean receiving ready-made antibodies, which is not what a vaccine injects, and the fact that the pathogen is weakened does not stop it from being immunogenic, which is the whole point of attenuation.

28. The natural pacemaker region that initiates each heartbeat is the:
- A) bundle of His
 - B) vagus nerve
 - C) sinoatrial (SA) node
 - D) atrioventricular (AV) node
 - E) Purkinje fibres

Answer: C

The sinoatrial node, a small region in the wall of the right atrium, depolarises fastest of all cardiac cells and initiates each heartbeat, making it the natural pacemaker. The atrioventricular node is only the normal relay to the ventricles, and the bundle of His and Purkinje fibres conduct the impulse onward; the vagus nerve modulates the rate but does not set the rhythm.

29. Gas exchange in the pulmonary alveoli is highly efficient mainly because:
- A) the alveoli are muscular and pump air
 - B) each alveolus contains cartilage rings
 - C) the alveoli store oxygen like a tank
 - D) their enormous total surface area is combined with an extremely thin (squamous) wall
 - E) alveolar air is under high pressure

Answer: D

Diffusion efficiency scales with surface area divided by wall thickness. Hundreds of millions of alveoli provide roughly 70 square metres of exchange surface behind walls of extremely thin squamous cells (about half a micrometre), which is the structural basis of their efficiency. Alveoli have no muscle to pump air, no cartilage rings (those are in the larger airways), and they store no oxygen.

30. ADH (vasopressin) increases water reabsorption in the kidney by:
- A) constricting the efferent arteriole
 - B) inserting aquaporin channels into the collecting duct wall
 - C) opening sodium channels in the loop of Henle
 - D) raising glomerular filtration pressure
 - E) dissolving urea in the filtrate

Answer: B

ADH binds V2 receptors on collecting duct cells and triggers insertion of aquaporin-2 channels into their membrane, so water leaves the filtrate down the medullary osmotic gradient and urine becomes concentrated. It does not act by changing filtration pressure, opening sodium channels, or constricting the efferent arteriole; it is a water-permeability hormone.

31. PCR (the polymerase chain reaction) requires a heat-stable DNA polymerase because:
- A) the enzyme must work at refrigerator temperature
 - B) DNA cannot be copied at low temperatures
 - C) the primers are destroyed by heat and must be resynthesised
 - D) heat is needed to insert the product into bacteria
 - E) each cycle includes a high-temperature step (about 95 °C) that separates the DNA strands

Answer: E

Each PCR cycle starts with a denaturation step near 95 degrees Celsius that separates the DNA strands, and this step repeats dozens of times. An ordinary polymerase would be inactivated after the first few cycles, so a heat-stable polymerase such as Taq (from hot-spring bacteria) is needed to survive the whole programme. It is the high-temperature strand-separation step, not low temperature, that demands the special enzyme.

32. An antibiotic blocks the bacterial 70S ribosome but not the eukaryotic 80S ribosome. This selectivity means the drug can:
- A) work only against viruses
 - B) act on the patient's mitochondria only
 - C) harm bacteria without directly stopping protein synthesis in the patient's own cells
 - D) be given at any dose without side effects
 - E) kill all cells equally

Answer: C

This is selective toxicity: because bacterial ribosomes are 70S while the patient's own cytoplasmic ribosomes are 80S, the drug can shut down bacterial protein synthesis without directly stopping protein synthesis in the patient's cells. It is not side-effect free at any dose, since human mitochondria carry their own 70S-like ribosomes and can be secondarily affected, and the drug targets bacteria, not viruses or mitochondria alone.

Section C - Chemistry (Q33-47)

33. What is the oxidation number of carbon in methanol, CH₃OH?
- A) -4 B) -2 C) 0 D) +2 E) +4

Answer: B

Oxidation numbers sum to zero in a neutral molecule. Assign hydrogen +1 (four atoms give +4) and oxygen -2: $C + 4 - 2 = 0$, so $C = -2$. The +4 value belongs to the carbon of CO₂ and the -4 value to methane; methanol sits between them on the reduction ladder at -2.

34. In the reaction $\text{Cl}_2 + 2\text{NaOH} \rightarrow \text{NaCl} + \text{NaOCl} + \text{H}_2\text{O}$, the chlorine:
- A) is simultaneously oxidised and reduced (disproportionation): $0 \rightarrow -1$ in NaCl and $0 \rightarrow +1$ in NaOCl
 - B) is reduced only
 - C) is oxidised only
 - D) keeps oxidation number 0 in both products
 - E) is unchanged, only dissolved

Answer: A

Follow the chlorine. In Cl₂ both atoms have oxidation number 0. In NaCl the chlorine ends at -1 (it gained an electron, so it was reduced); in NaOCl it ends at +1 (it lost an electron to oxygen, so it was oxidised). The same element is simultaneously oxidised and reduced: a disproportionation, and the arithmetic balances because one atom goes down 1 and the other up 1.

35. Which molecule has zero net dipole moment despite having polar bonds?

- A) NH_3 B) SO_2 C) H_2O D) BF_3 E) CHCl_3

Answer: D

BF_3 is trigonal planar: three identical polar B-F bonds point symmetrically at 120 degrees, so their dipoles cancel vectorially and the molecule has zero net dipole despite the polar bonds. In NH_3 (trigonal pyramidal), H_2O and SO_2 (both bent) and CHCl_3 (an asymmetric tetrahedron) the bond dipoles do not cancel, leaving a permanent net dipole.

36. Which molecule has a trigonal pyramidal shape?

- A) H_2O B) NH_3 C) BF_3 D) CO_2 E) CH_4

Answer: B

NH_3 has three bonding pairs plus one lone pair on nitrogen. The tetrahedral electron arrangement leaves the molecular shape trigonal pyramidal with a bond angle of about 107 degrees. CH_4 (no lone pairs) is tetrahedral, H_2O (two lone pairs) is bent, BF_3 is trigonal planar and CO_2 is linear.

37. What is the pH of a 0.005 M solution of $\text{Ba}(\text{OH})_2$ at 25 °C? ($\text{Ba}(\text{OH})_2$ releases 2 OH^- per formula unit; $K_w = 1.0 \times 10^{-14}$)

- A) 1.7 B) 2 C) 11 D) 12 E) 13

Answer: D

Step 1: each $\text{Ba}(\text{OH})_2$ releases two hydroxide ions, so $[\text{OH}^-] = 2 \times 0.005 = 0.010 \text{ M}$.

Step 2: $\text{pOH} = -\log(0.010) = 2$

Step 3: $\text{pH} = 14 - 2 = 12$ at 25 degrees Celsius

Forgetting the factor of 2 gives $[\text{OH}^-] = 0.005 \text{ M}$, pOH of about 2.3 and pH about 11.7, which is how the tempting value near 11 arises; the value 2 is the pOH , not the pH .

38. Solution A has pH 4 and solution B has pH 6. The $[\text{H}^+]$ of A compared with B is:

- A) 1.5 times greater
B) 2 times greater
C) 10 times greater
D) 100 times greater
E) 1000 times greater

Answer: D

pH is a logarithmic scale: $[\text{H}^+]$ of A is 10^{-4} M and of B is 10^{-6} M .

The ratio is $10^{-4} / 10^{-6} = 10^2 = 100$, so A is 100 times more acidic

The difference of 2 pH units is a factor of 100 in concentration, not a factor of 2.

39. How many atoms (total) are present in 0.25 mol of CO_2 ? ($N_a = 6.02 \times 10^{23} \text{ mol}^{-1}$)

- A) 0.75×10^{23} B) 1.5×10^{23} C) 3.0×10^{23} D) 4.5×10^{23} E) 6.0×10^{23}

Answer: D

Step 1: number of CO_2 molecules = $0.25 \text{ mol} \times 6.02 \times 10^{23} \text{ mol}^{-1} = 1.5 \times 10^{23}$

Step 2: each molecule contains 3 atoms (1 carbon + 2 oxygens), so total atoms = $3 \times 1.5 \times 10^{23} = 4.5 \times 10^{23}$. Stopping after step 1 and reporting 1.5×10^{23} is the classic error of counting molecules instead of atoms.

40. Heating 10.0 g of pure CaCO_3 ($M = 100 \text{ g/mol}$) would release 2.24 L of CO_2 at STP. An experiment on an impure sample of the same mass releases 1.68 L at STP. The percentage of CaCO_3 in the sample is:

- A)** 25% **B)** 60% **C)** 67% **D)** 75% **E)** 133%

Answer: D

The CO_2 volume is proportional to the amount of CaCO_3 present, and the same mass of pure CaCO_3 would release 2.24 L.

So the percentage of CaCO_3 in the sample is $1.68 / 2.24 \times 100 = 75\%$

Taking the deficit instead ($0.56 / 2.24 = 25\%$) measures the impurity rather than the carbonate, and dividing 0.56 by the actual yield 1.68 gives a meaningless 33%.

41. A gas at 27°C and 1.0 atm is sealed in a rigid vessel and heated to 127°C . The new pressure is closest to:

- A)** 0.75 atm **B)** 1.0 atm **C)** 1.11 atm **D)** 1.33 atm **E)** 4.2 atm

Answer: D

Gay-Lussac's law (constant volume, constant amount): $P_2 = P_1 \times T_2 / T_1$, with absolute temperatures.

$T_1 = 27 + 273 = 300 \text{ K}$, $T_2 = 127 + 273 = 400 \text{ K}$

$P_2 = 1.0 \times 400/300 = 1.33 \text{ atm}$

Working directly in Celsius (ratio $127/27$) is meaningless because the Celsius zero is arbitrary, and inverting the ratio gives the impossible result that heating lowers the pressure.

42. A chlorine atom is smaller than a sodium atom mainly because:

- A)** chlorine has more protons pulling on electrons in the same shell
B) chlorine gains electrons, sodium loses them
C) sodium is a metal and chlorine is a gas
D) chlorine's electrons repel each other more strongly
E) chlorine has more electron shells

Answer: A

Sodium and chlorine belong to the same period, so both have their outer electrons in the same third shell. Chlorine's nucleus carries 17 protons against sodium's 11, and this greater nuclear charge pulls the same-shell electrons closer, making the atom smaller. The comparison concerns neutral atoms, so ion formation is irrelevant, and the shell count is identical, not higher, for chlorine.

43. Methanol reacts with propanoic acid under reflux with concentrated H_2SO_4 . The organic product is:

- A)** propanal
B) methanoic acid
C) propanoic anhydride
D) propan-1-ol
E) methyl propanoate

Answer: E

An alcohol and a carboxylic acid under reflux with concentrated sulfuric acid undergo esterification. The alcohol supplies the alkyl part (methanol gives methyl) and the acid supplies the carboxylate part (propanoic acid gives propanoate), so the product is methyl propanoate, $\text{CH}_3\text{CH}_2\text{COOCH}_3$, with water as by-product. The reverse name (propanyl-type esters) or any aldehyde or anhydride misreads which fragment comes from which reagent.

44. Hydrogen bromide adds to propene ($\text{CH}_3\text{-CH=CH}_2$). The major product is:

- A) propane
- B) 1,2-dibromopropane
- C) 2-bromopropane
- D) 2-bromopropene
- E) 1-bromopropane

Answer: C

Hydrogen bromide adds to propene by Markovnikov's rule: H attaches to the double-bond carbon that already has more hydrogens (the terminal CH_2), because that route passes through the more stable secondary carbocation, and Br then bonds to the middle carbon. The product is 2-bromopropane, $\text{CH}_3\text{-CHBr-CH}_3$. The bromine on the end carbon (1-bromopropane) would be the anti-Markovnikov product, obtained only in the presence of peroxides, and the dibromide would need Br_2 rather than HBr .

45. For the heterogeneous equilibrium $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$, the equilibrium constant is written as:

- A) $K_c = [\text{CaO}][\text{CO}_2]$
- B) $K_c = 1$, since solids are constant
- C) $K_c = [\text{CO}_2]$
- D) $K_c = 1/[\text{CO}_2]$
- E) $K_c = [\text{CO}_2]/([\text{CaO}][\text{CaCO}_3])$

Answer: C

In a heterogeneous equilibrium the pure solids appear in fixed amounts, so their terms are constants that are folded into the equilibrium constant and omitted from the expression.

Only the gaseous product remains: $K_c = [\text{CO}_2]$

Including the solids in a quotient, or inverting the expression, misapplies the rule that pure phases are excluded.

46. In a working galvanic (voltaic) cell:

- A) anions in the salt bridge migrate toward the cathode compartment
- B) electrons flow from cathode to anode through the external wire
- C) oxidation occurs at the cathode
- D) electrons flow through the salt bridge
- E) oxidation occurs at the anode and the electrons released flow through the external wire from anode to cathode

Answer: E

In any working galvanic cell, oxidation occurs at the anode, and the electrons released travel through the external wire from anode to cathode, where reduction takes place. The salt bridge carries ions, never electrons, and its anions migrate toward the anode compartment to neutralise the positive charge building up there (cations drift toward the cathode side). Any statement that puts oxidation at the cathode, or electrons in the bridge, contradicts these roles.

47. Four 0.10 M solutions are prepared: glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), NaCl, CaCl_2 , and sucrose. Which has the lowest freezing point?

- A) CaCl_2 (three ions per formula unit)
- B) sucrose (no ions)
- C) NaCl (two ions)
- D) all four freeze at the same temperature
- E) glucose (no ions)

Answer: A

Freezing-point depression is a colligative property: it depends on the number of dissolved particles, not their identity. Per litre of 0.10 M solution: CaCl_2 gives about 0.30 mol of ions (3 per formula unit), NaCl about 0.20 mol (2 per unit), while glucose and sucrose give only 0.10 mol of molecules each. The greatest particle count produces the greatest depression, so the CaCl_2 solution freezes lowest, and identical freezing points would require identical particle counts, which is not the case here.

Section D - Mathematics (Q48-54)

48. Solve the inequality $8 - 5x < -12$.

- A) $x < -4$ B) $x > 4$ C) $x < 4$ D) $x \geq 4$ E) $x > -4$

Answer: B

Step 1: subtract 8 from both sides: $-5x < -20$. Step 2: divide both sides by -5 , which flips the direction of the inequality: $x > 4$. Check $x = 5$: $8 - 25 = -17$, which is indeed less than -12 . The slips to avoid are forgetting the flip (giving $x < 4$) and losing the strictness (giving x at least 4, although $x = 4$ itself gives $8 - 20 = -12$, which fails the strict inequality).

49. Solve $-3 < 2x + 1 \leq 7$.

- A) $-4 < x \leq 6$ B) $-2 \leq x \leq 3$ C) $-2 < x < 3$ D) $-2 < x \leq 3$ E) $-1 < x \leq 4$

Answer: D

Step 1: subtract 1 from all three parts: $-4 < 2x \leq 6$. Step 2: divide by the positive number 2 (signs and strictness unchanged): $-2 < x \leq 3$. The left bound stays strict and the right bound stays inclusive, exactly mirroring the original statement; making either bound strict or both inclusive misreads which comparison applied to which side.

50. If $2^x = 40$, then x satisfies:

- A) $x = 6$ B) $4 < x < 5$ C) $6 < x < 7$ D) $x = 5$ E) $5 < x < 6$

Answer: E

Bracket 40 between consecutive powers of 2: $2^5 = 32$ and $2^6 = 64$

Since 2^x increases continuously with x , 40 being between 32 and 64 forces $5 < x < 6$ (the exact value is $\log_2 40$, about 5.32). The equality answers are impossible because neither 32 nor 64 equals 40.

51. One card is drawn at random from a standard 52-card deck. What is the probability it is a heart or a king?

- A)** $1/4$ **B)** $1/13$ **C)** $4/13$ **D)** $13/52$ **E)** $17/52$

Answer: C

Use inclusion-exclusion on the overlap: 13 hearts + 4 kings – 1 king of hearts (counted in both sets) = 16 favourable cards.

Probability = $16/52 = 4/13$

Adding $13 + 4 = 17$ without removing the overlap overcounts by exactly one card, and $13/52$ would cover only the hearts.

52. A straight ramp 8 m long makes a 30° angle with the horizontal. How high is its upper end above the ground? ($\sin 30^\circ = 0.5$, $\cos 30^\circ = 0.87$)

- A)** 2 m **B)** 4 m **C)** 6.9 m **D)** 8 m **E)** 16 m

Answer: B

The height is the side opposite the 30° angle in a right triangle with the ramp as hypotenuse: height = $8 \times \sin 30^\circ = 8 \times 0.5 = 4$ m. Using $\cos 30^\circ$ instead gives 8×0.87 , which is about 6.9 m, but that is the horizontal projection of the ramp along the ground, not the vertical rise.

53. If $f(x) = 2x - 1$, then $f(f(x))$ equals:

- A)** $2x - 3$ **B)** $3x - 4$ **C)** $4x - 3$ **D)** $4x - 1$ **E)** $4x + 1$

Answer: C

Substitute the inner function into the outer one: $f(f(x)) = 2(2x - 1) - 1 = 4x - 2 - 1 = 4x - 3$. The common slip of multiplying by 2 only once produces $2x - 3$, but composition applies the doubling and the subtraction at both stages.

54. The circle $(x - 1)^2 + (y + 2)^2 = 16$ has centre and radius:

- A)** (1, -2), radius 4
B) (1, -2), radius 16
C) (-1, 2), radius 4
D) (-1, 2), radius 16
E) (1, 2), radius 4

Answer: A

Compare with the standard form $(x - h)^2 + (y - k)^2 = r^2$

Here $h = 1$ (so $x - 1$) and $k = -2$, since $y + 2$ equals $y - (-2)$, giving centre (1, -2). The right side is $r^2 = 16$, so the radius is $r = \sqrt{16} = 4$, not 16 itself; reading the signs of h and k straight from the printed plus signs instead gives the wrong quadrant.

Section E - Physics (Q55-60)

55. A 24 V battery drives a $2\ \Omega$ resistor in series with a parallel pair of $12\ \Omega$ and $6\ \Omega$. What current flows from the battery?

- A)** 2 A **B)** 3 A **C)** 4 A **D)** 6 A **E)** 12 A

Answer: C

Step 1: reduce the parallel pair: $1/R = 1/12 + 1/6 = 1/12 + 2/12 = 3/12$, so $R = 4$ ohms (equivalently $12 \times 6 / 18 = 4$). Step 2: in series with the 2 ohm resistor the total is $2 + 4 = 6$ ohms.

Step 3: from Ohm's law, $I = V/R = 24/6 = 4$ A

Reporting $24/12 = 2$ A would ignore the parallel section, and $24/4 = 6$ A would ignore the series resistor.

56. A 60 W lamp is designed for a 120 V supply. The resistance of its filament is:

- A) 0.5 Ω B) 2 Ω C) 60 Ω D) 240 Ω E) 7200 Ω

Answer: D

Use $P = V^2 / R$ and rearrange: $R = V^2 / P = 120^2 / 60 = 14400 / 60 = 240$ ohms

The value 2 ohms comes from computing V/P , which is actually the current (2 A) that the lamp draws, not its resistance.

57. Water boiling at atmospheric pressure stays at exactly 100 °C even though the flame keeps supplying heat. This is because:

- A) the supplied energy goes into the latent heat of vaporisation, breaking intermolecular attractions instead of raising temperature
B) boiling only occurs without heating
C) the thermometer is faulty above 90 °C
D) water has no specific heat at 100 °C
E) the molecules stop moving at 100 °C

Answer: A

Boiling is a phase change: all the heat supplied goes into the latent heat of vaporisation (about 2260 J per gram), the energy needed to pull molecules apart against their intermolecular attractions, so none of it raises the temperature until the last liquid has evaporated. A phase change in progress is isothermal, which is why the reading pins at 100 degrees Celsius; the molecules certainly do not stop moving, and boiling requires continued heat input.

58. Cars once had rigid metal bumpers; modern cars crumple in a crash. The physics reason a crumple zone reduces injury is that it:

- A) eliminates the change of momentum
B) extends the collision time, reducing the average force for the same change of momentum
C) converts the passenger into the system's centre of mass
D) reduces the car's momentum to zero instantly
E) increases the impulse delivered

Answer: B

A crash fixes the change of momentum the car must undergo; that change of momentum is the impulse, $F \times \Delta t$. A crumple zone extends the collision time Δt by deforming, so for the same Δp the average force F on the occupants is smaller, and injury scales with force. The momentum change cannot be eliminated or made instant without harm, and the impulse delivered is set by the crash, not increased by the crumple zone.

59. On a distance-time graph, a horizontal segment means the object is:

- A) moving at constant non-zero speed
B) decelerating to rest at the end of the segment
C) moving backwards
D) accelerating
E) stationary (zero speed for that interval)

Answer: E

On a distance-time graph the gradient of the line is the speed. A horizontal segment has zero gradient, so the distance does not change during that interval: the object is stationary. The same flat line on a velocity-time graph would mean constant non-zero speed, which is exactly why reading the axis label before interpreting matters.

60. A block of density 0.90 g/cm^3 floats in water (1.00 g/cm^3). What fraction of the block's volume is submerged?

- A)** 10% **B)** 25% **C)** 50% **D)** 90% **E)** 100%

Answer: D

Floating equilibrium: weight equals upthrust, so $\rho_{\text{block}} \times V \times g = \rho_{\text{water}} \times V_{\text{submerged}} \times g$. The g and V cancel, giving $V_{\text{submerged}} / V = 0.90 / 1.00 = 0.90$, so 90% of the block sits below the surface. The 10% figure describes the fraction above water, the complement of the answer.